FTM iMCQ3
Fall 2025

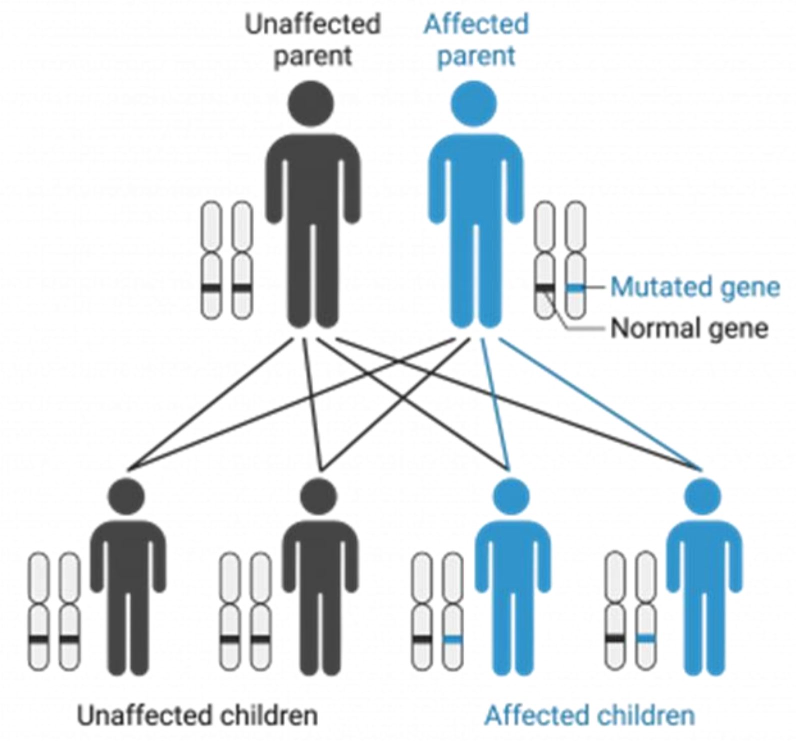
Genetics and Biochemistry

KEY File

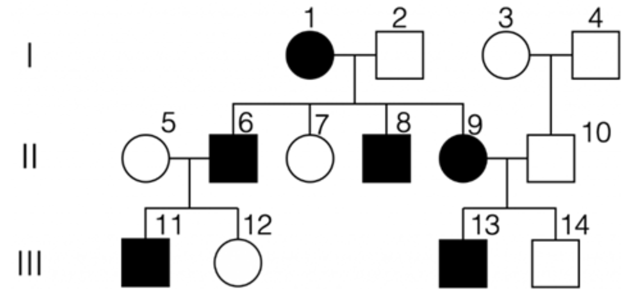
FTM IMCQ3

Genetics

A couple wanting to start a family comes to an appointment with a genetic counselor. One of the parents is affected by a genetic disorder. The counselor explains the recurrence risk as shown in the figure. Which of the following is the most likely genetic disorder in this family?



A. Alkaptonuria
B. Marfan syndrome
C. Duchenne muscular dystrophy
D. Klinefelter syndrome
E. Turner syndrome
F. Hemophilia A



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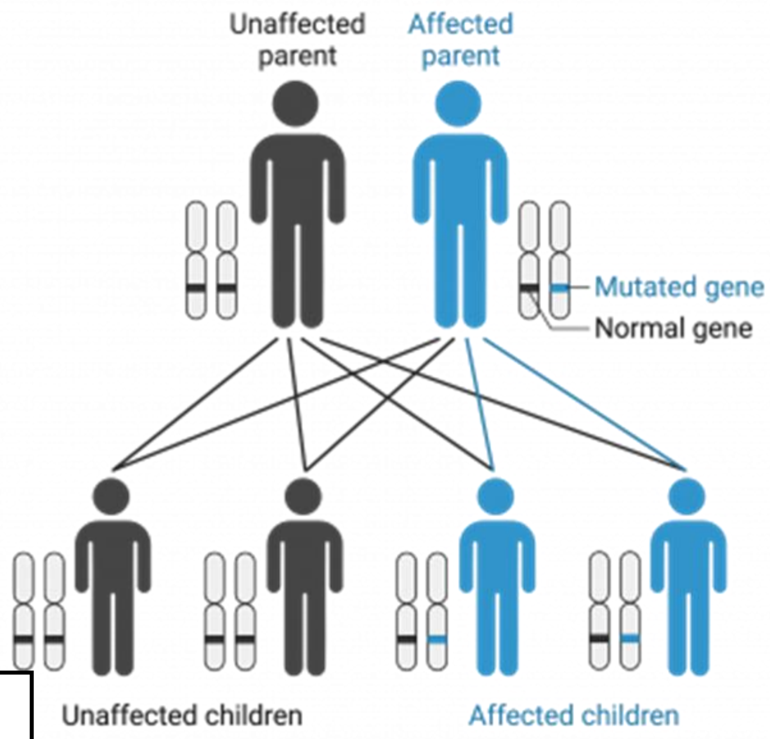

SOM.MK.III.BPM1.1.FTM.3.GNET.0021From a description of a pedigree, identify autosomal dominant (AD) and autosomal recessive (AR) inheritance; Compare and contrast Mendelian and non-Mendelian genetics
SOM.MK.III.BPM1.1.FTM.3.GNET.0028Demonstrate how disease liability is transmitted for autosomal dominant and autosomal recessive disorders, beginning from the first "de novo" occurrence.

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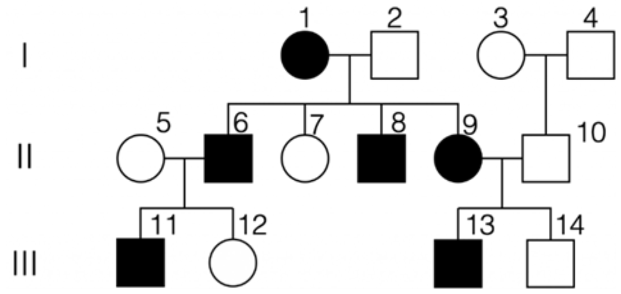

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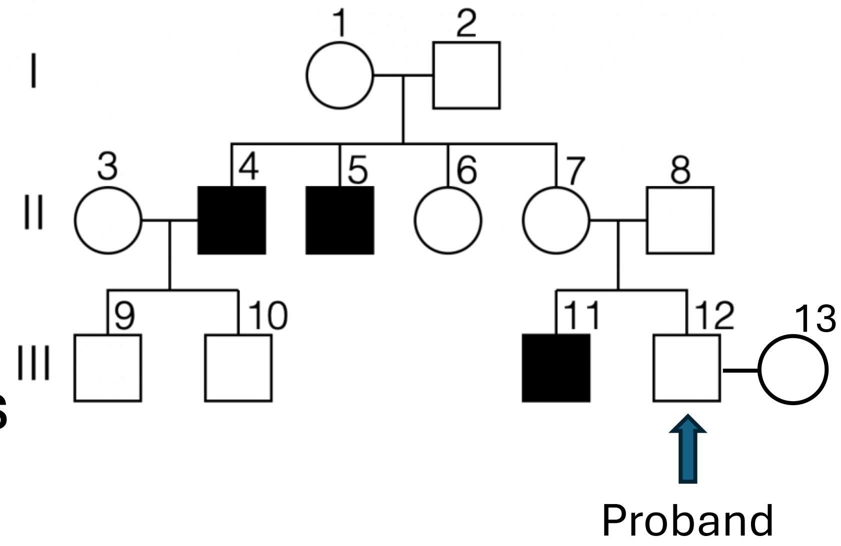
A couple wanting to start a family comes to an appointment with a genetic counselor (III-12 and III-13). The father (III-12) has a family history of a genetic disorder. The counselor provides a pedigree analysis shown in the figure. Which of the following best predicts the risk that the couple will have a child affected with the genetic disorder affecting this family?



A. 1/2
B. 2/3
C. Almost nil
D. ¼ chance
E. 100% chance

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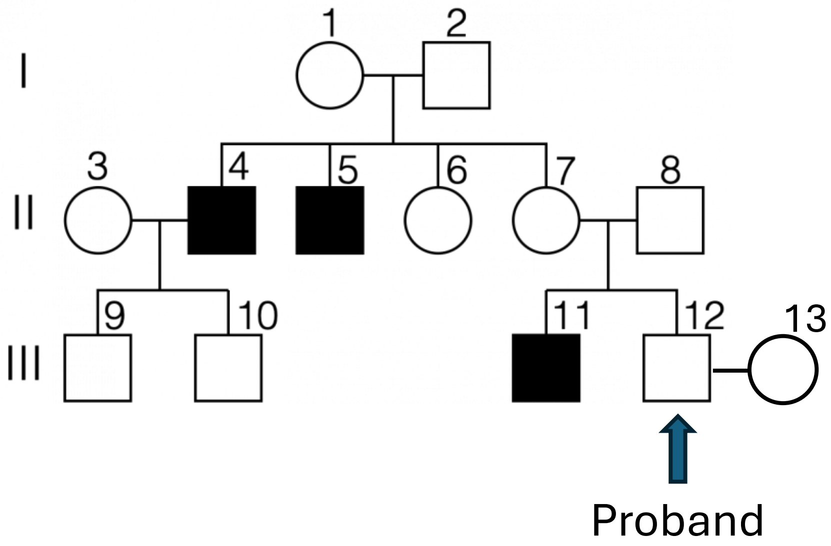

A. 1/2

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✔ C. Almost nil

D. ¼ chance

E. 100% chance

SOM.MK.III.BPM1.1.FTM.3.GNET.0049 Distinguish the different characteristics of recessive and dominant X-linked traits using pedigrees
SOM.MK.III.BPM1.1.FTM.3.GNET.0048 Demonstrate how disease liability is transmitted for autosomal dominant diseases,
autosomal recessive diseases and X-linked diseases
SOM.MK.III.BPM1.1.FTM.3.GNET.0050 Identify obligate carriers in a pedigree depicting an X-linked trait

A 4-year-old boy is brought to the physician by his parents because of increased sensitivity to sunlight. His parents say that he loves playing outside but has sunburned easily since infancy. Physical examination shows dry, rough facial skin with freckling. Family history includes a sister with many freckles who loves gaming and does not play outside, and a paternal uncle with cutaneous squamous cell carcinoma, as shown. A diagnosis of xeroderma pigmentosum is made. Which of the following genetic phenomena is seen in this family?

Patient

Uncle

Sister

A. Variable expression
B. Autosomal dominant inheritance
C. X-linked recessive
D. Mitochondrial inheritance
E. Incomplete penetrance
F. Anticipation

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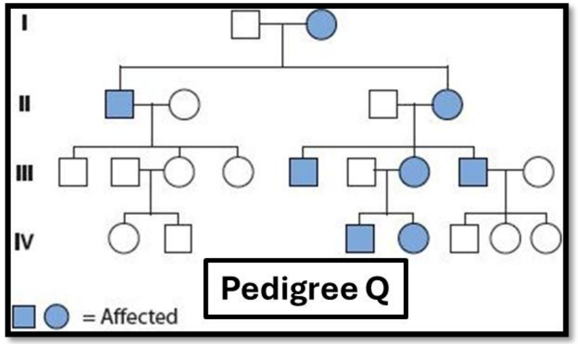
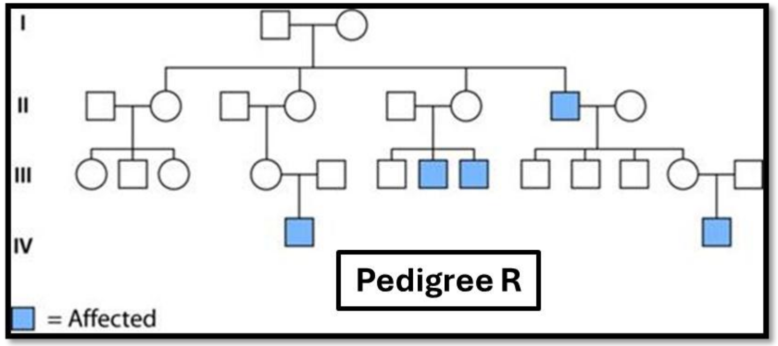
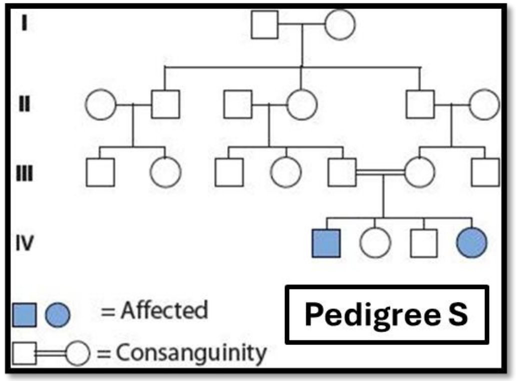
SOM.MK.III.BPM1.1.FTM.3.GNET.0043 Compare and contrast variable expressivity and incomplete penetrance; Give examples of each.

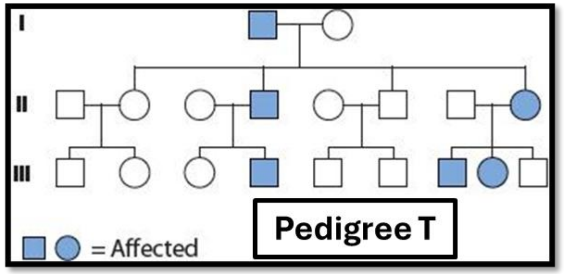
A 2-week-old female newborn is brought to the physician for a follow-up examination because of a positive newborn screening test. Newborn screening tests show elevated blood galactose concentration which was later confirmed by enzyme assays. Which of the following best represents the patient's family history?

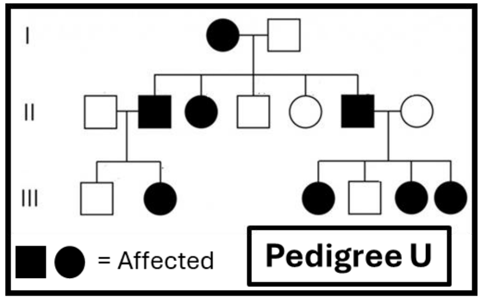
A. Pedigree Q
B. Pedigree R
C. Pedigree S
D. Pedigree T
E. Pedigree U











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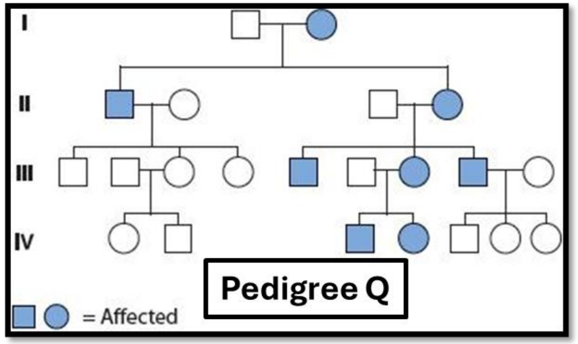
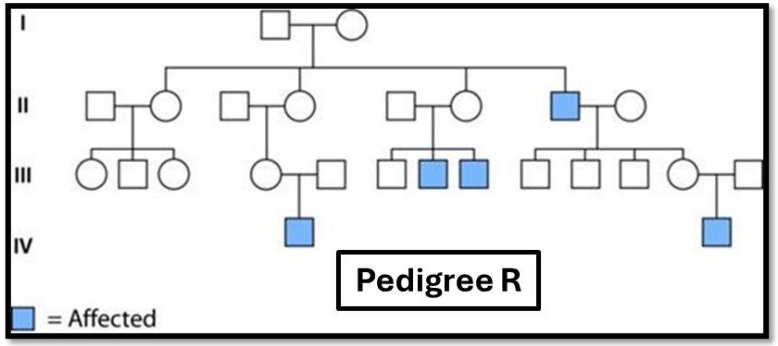
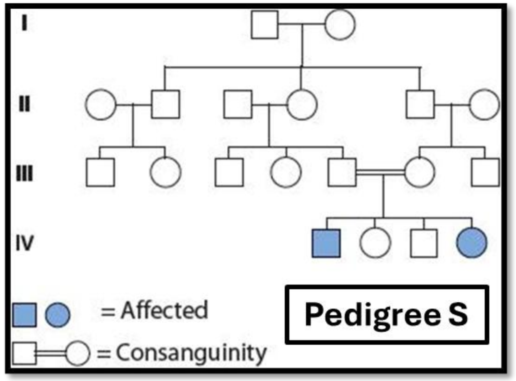
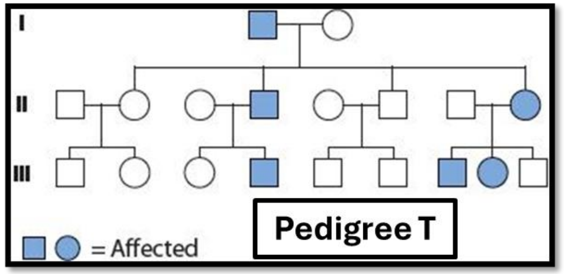
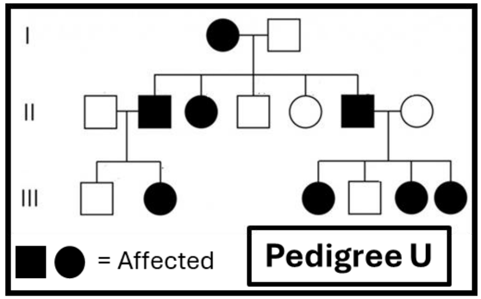




SOM.MK.III.BPM1.1.FTM.3.GNET.0032Distinguish the mode of inheritance, identify the main clinical features, and interpret genetic principles illustrated by the following following autosomal recessive disorders: CF, PKU, Tay-Sachs disease, Sickle cell disease, thalassemia, hemochromatosis, homocystineuria, SCID due to ADA deficiency, alkaptonuria







A 2-year-old boy is brought to the physician by his mother because of a 2-day history of lip lacerations. Physical examination shows lacerations of the lower lip and lacerations with crusting and scarring of the left thumb. A diagnosis of HGPRT enzyme deficiency was made. During discussions of his management, his mother says that she is currently 18 weeks pregnant with her second son and has concerns of him being affected with this disorder. Which of the following best represents the risk of her fetus being affected by this disorder?

A. 25%
B. 33%
C. 50%
D. 100%
E. 0 (zero)

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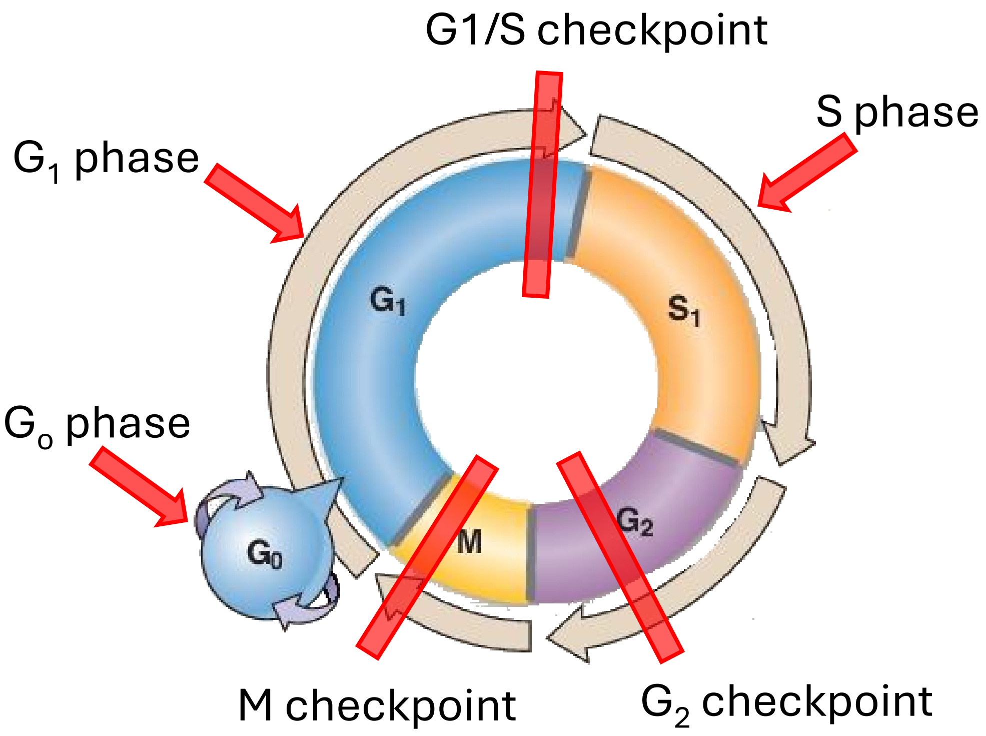
B. 33%
C. 50%
D. 100%
E. 0 (zero)

SOM.MK.III.BPM1.1.FTM.3.GNET.0037Distinguish the mode of inheritance, identify the main clinical features, and interpret genetic principles illustrated by the following X-linked disorders: Hemophilia A and B, G6PD deficiency, Duchenne and Becker muscular dystrophy, red/green color blindness. Lesch Nyhan syndrome, X-SCID as examples of X-linked disorders

An investigator is studying the cellular response to DNA damage in human fibroblasts by exposing the cells to ionizing radiation. An increase in p53 protein levels is observed shortly afterwards. This increase leads to cell cycle arrest allowing time for DNA repair. Which of the following points of the cell cycle is this cell cycle arrest most likely to be observed?



A. G_0
B. Early G_1 phase
C. G1/S checkpoint
D. S phase
E. G_2 checkpoint
F. M checkpoint

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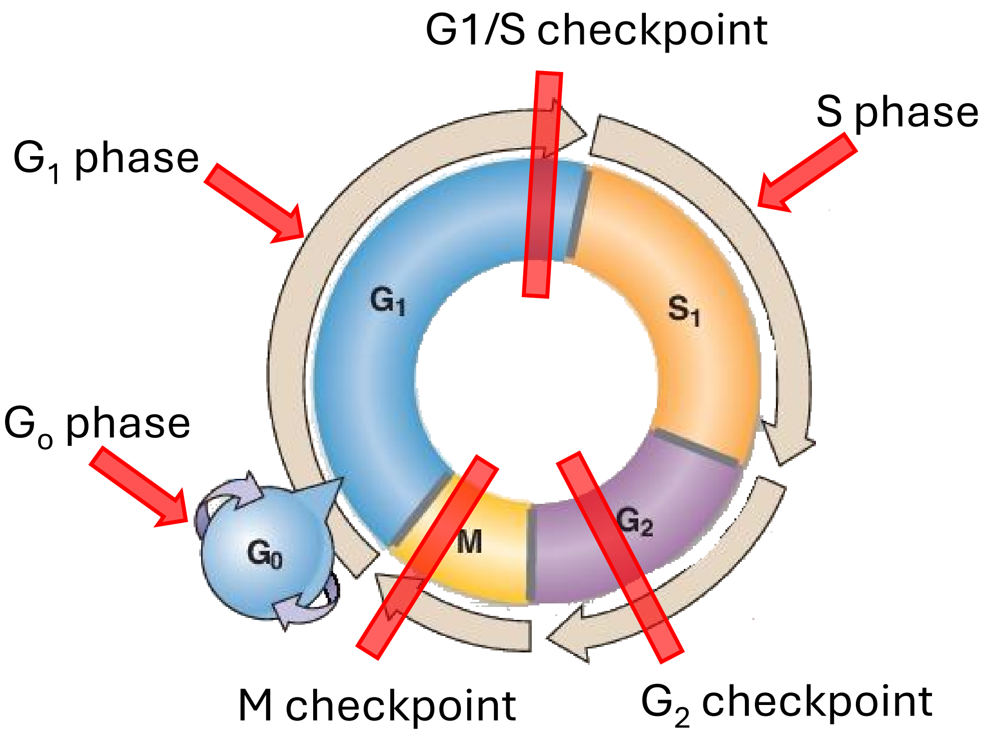

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B. Early G_1 phase

✓ C. G1/S checkpoint

D. S phase

E. G_2 checkpoint

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SOM.MKIII.BPM1.1.FTM.3.GNET.0003
Indicate the role of p53 in the cell cycle: Role in genome surveillance and its role in initiation of apoptosis

A reproductive biologist working at a fertility clinic is analyzing ovarian tissue samples from a healthy adult female patient undergoing evaluation for oocyte preservation. The investigator notes that the majority of the oocytes present in the ovarian follicles are primary oocytes. These cells have been arrested in a specific phase of meiosis since fetal development. Which of the following phases of meiosis are these cells arrested?

A. Prophase I
B. Prophase II
C. Metaphase I
D. Metaphase II
E. Anaphase I
F. Anaphase II

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SOM.MKIII.BPM1.1.FTM.3.GNET.0005
Explain the significance of mitosis, summarize the events, and analyze structures of mitosis including Prophase: Events in the nucleus, formation of spindle in cytoplasm, Metaphase: Association of chromosomes with spindle fibers, Anaphase: Separation of sister chromatids, Telophase: Cytokinesis, re-formation of nuclei

A 6-month-old infant is brought to the physician by the parents for evaluation of developmental delay and hypotonia. Physical examination shows upslanting palpebral fissures, a single palmar crease, and a protruding tongue. Chromosomal analysis shows two distinct cell lines: one with a chromosome complement of 46,XX and another with 47,XX,+21. Which of the following genetic mechanisms has most likely occurred to explain this patient's condition?

A. Uniparental disomy of chromosome 21
B. Mitotic non-disjunction during embryogenesis
C. Non-disjunction during meiosis I
D. Non-disjunction during meiosis II
E. Non-homologous recombination during prophase I

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SOM.MKIII.BPM1.1.FTM.3.GNET.0013
Compare and contrast the outcome of nondisjunction that might occur during meiosis or in mitosis

An investigator is reviewing the human genome in regions of DNA that have been very difficult to sequence due to very dense packing. A new technique now allows for accurate sequencing. The investigator identifies a DNA sequence that closely resembles a known functional gene but contains multiple mutations that prevent transcription. Which of the following most likely describes this newly identified region of DNA?

A. SINES
B. LINES
C. SNPs
D. Pseudogene
E. Low copy repeat

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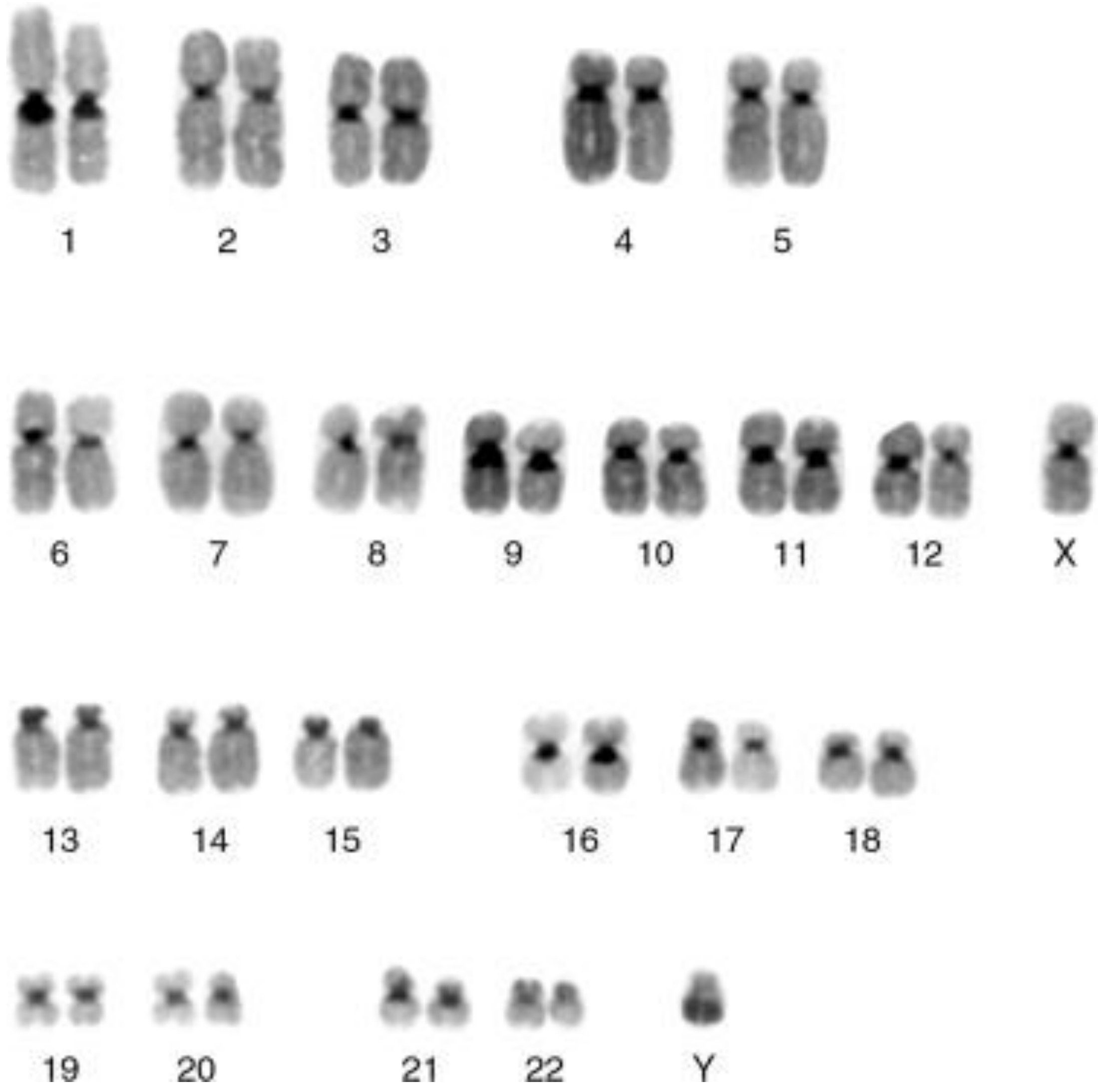
SOM.MKIII.BPM1.1.FTM.3.GNET.0016 Compare and contrast different types of repetitive DNA including low copy repeats, SSR, VNTR, LINE, SINE, pseudogenes, and gene families

An investigator is performing C-banding karyotype analysis which stains heterochromatin around the centromeres as shown in the image. The investigator is interested in identifying acrocentric chromosomes. Which of the following chromosomes most likely has this morphology?



A. Chromosome 1
B. Chromosome 5
C. Chromosome 13
D. Chromosome 19
E. X chromosome

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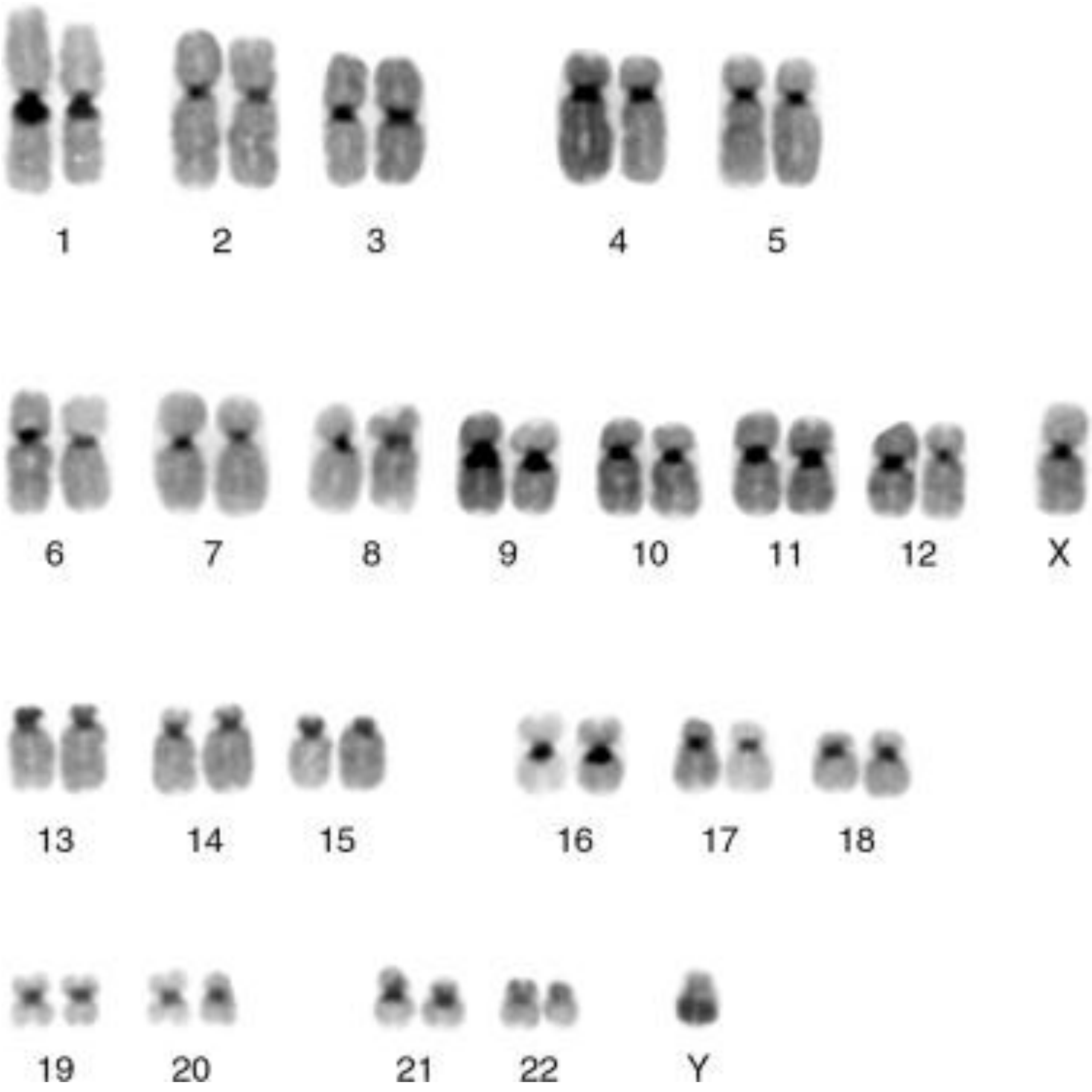

A. Chromosome 1
B. Chromosome 5
✔ C. Chromosome 13
D. Chromosome 19
E. X chromosome

SOM.MKIII.BPM1.1.FTM.3.GNET.0018 Discuss the structure of chromosomes in the human (male and female), and general terminology used to describe karyotypes; Compare nuclear DNA to organellar DNA (such as the mitochondrial)
SOM.MKIII.BPM1.1.FTM.3.GNET.0011 Describe the distinguishing features and the morphological classification of chromosomes including metacentric, submetacentric, acrocentric, and telocentric

An investigator is examining buccal epithelial cells under a microscope of a patient with Klinefelter syndrome. They notice a small, densely stained region close to the nuclear membrane in all the cells. Which of the following best explains the condensed structure in the cells?

A. Lyonization of the Y chromosome
B. Formation of a nucleolus
C. Mitochondrial DNA aggregation
D. Condensation of autosomal heterochromatin
E. X-chromosome inactivation

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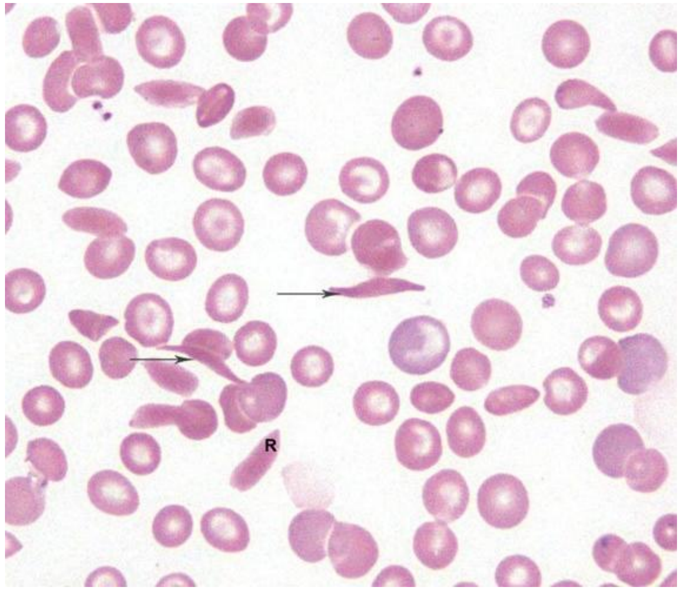
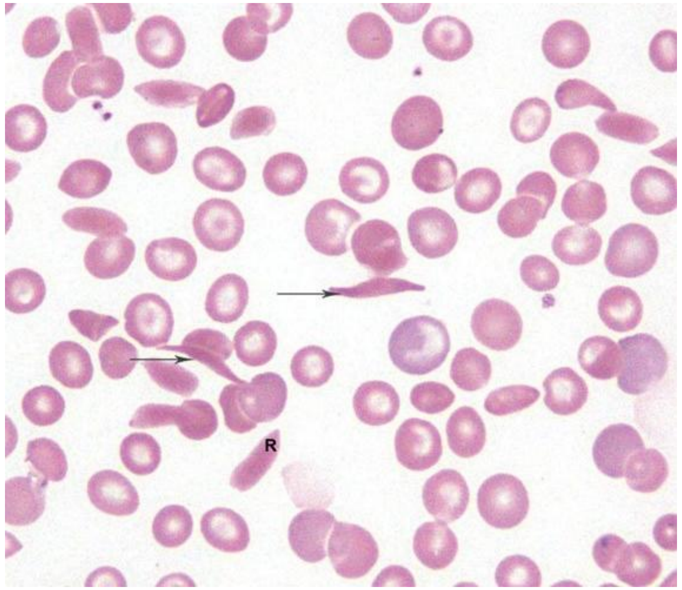
SOM.MKIII.BPM1.1.FTM.3.GNET.0105 Explain the normal function of epigenetic gene silencing in development; including the role of X-inactivation

A 2-year-old boy is brought to the physician by his parents because of a 1-day history of painful swelling of the distal interphalangeal joints and generalized body pain. Physical examination shows yellowing of his skin (jaundice) and mucosal surfaces and an enlarged spleen. He is also shorter compared to the average children of his age. His pulse is 120/min, respirations are 18/min, and temperature is 100.4°F (38°C). Laboratory studies show: Hemoglobin: 7 g/dl (11-15). A peripheral blood smear is shown. Which of the following best describes the inheritance of this disorder and risk to future siblings?



A. His father is a carrier; 50% of his male siblings may be affected
B. His mother is a carrier; 50% of his female siblings may be affected
C. Both parents are carriers; 50% risk of siblings being affected
D. One parent is affected; 25% of his siblings may inherit this disease
E. Both parents are carriers; 25% risk of inheriting the disorder

A 2-year-old boy is brought to the physician by his parents because of a 1-day history of painful swelling of the distal interphalangeal joints and generalized body pain. Physical examination shows yellowing of his skin (jaundice) and mucosal surfaces and an enlarged spleen. He is also shorter compared to the average children of his age. His pulse is 120/min, respirations are 18/min, and temperature is 100.4°F (38°C). Laboratory studies show: Hemoglobin: 7 g/dl (11-15). A peripheral blood smear is shown. Which of the following best describes the inheritance of this disorder and risk to future siblings?



SOM.MK.III.BPM1.1.FTM.3.GNET.0032Distinguish the mode of inheritance, identify the main clinical features, and interpret genetic principles illustrated by the following following autosomal recessive disorders: CF, PKU, Tay-Sachs disease, Sickle cell disease, thalassemia, hemochromatosis, homocystineuria, SCID due to ADA deficiency, alkaptonuria

A. His father is a carrier; 50% of his male siblings may be affected
B. His mother is a carrier; 50% of his female siblings may be affected
C. Both parents are carriers; 50% risk of siblings being affected
D. One parent is affected; 25% of his siblings may inherit this disease
✓E. Both parents are carriers; 25% risk of inheriting the disorder

FTM IMCQ3

Biochemistry

A 28-year-old man comes to the physician because of a 3-month history of memory loss and following comments by friends who notice changes in his behaviour. Genetic analysis shows a mutation in the PrP gene, and a diagnosis of variant Creutzfeldt-Jakob disease is made due to changes in the prion protein. Which of the following is the most likely observed in this protein in this patient?

A. Formation of additional disulfide bridges
B. Conversion of α-helices to β-sheets
C. Conversion of β-sheets to α-helices
D. Disruption of hydrogen bonding
E. Disruption of electrostatic interactions
F. Formation of triple helices
G. Increased glycation of serine residues

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SOM.MK.I.BPM1.1.FTM.3.BCHM.0091; Discuss the change of the protein secondary structure in Prion disease.

A. Formation of additional disulfide bridges
B. Conversion of α-helices to β-sheets
C. Conversion of β-sheets to α-helices
D. Disruption of hydrogen bonding
E. Disruption of electrostatic interactions
F. Formation of triple helices
G. Increased glycation of serine residues

A 22-year-old woman comes to the physician because of a 3-day history of chest pain and shortness of breath. She has had a cough productive of green, foul-smelling sputum during this time. Her temperature is 38.7°C (101.7°F), pulse is 104/min, respirations are 20/min, and blood pressure is 138/81 mm Hg. Physical examination shows distant breath sounds and crackles in the lower lung field bilaterally. A chest x-ray shows bilateral patchy shadows at the lung bases. Sputum culture grows *Streptococcus pneumoniae* isolates. A therapeutic agent that increases positive supercoils during DNA replication is started. Which of the following pharmacologic agents was most likely prescribed for this patient?

A. Actinomycin D
B. Chloramphenicol
C. Camptothecin
D. Tetracycline
E. Quinolone
F. Etoposide
G. Streptomycin
H. Cytarabine

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G. Streptomycin
H. Cytarabine

SOM.MK.I.BPM1.1.FTM.3.BCHM.0069 Explain the mode of action of common antibiotics that interfere with translation: Initiation inhibitors (streptomycin), Elongation inhibitors (Tetracycline, chloramphenicol, erythromycin, puromycin, cycloheximide).

A 14-month-old girl is brought to the physician because of a 6-month history of worsening motor incoordination and mental obtundation. On examination, she has retinal pallor with a prominent macula. Flaccid paralysis developed at 1 year of age. Targeted gene mutation analysis shows the findings in the figure. Which of the following mutations most likely occurred?

. . . – Arg – Ile – Ser – Tyr – Gly – Pro – Asp – . . .
. . . CGT ATA TCC TAT GCC CCT GAC

. . . CGT ATA TCT ATC CTA TGC CCC TGA C . . .
. . . – Arg – Ile – Ser – Ile – Leu – Cys – Pro – Stop

Altered reading frame

A. Point mutation
B. Nonsense mutation
C. Frame-shift mutation
D. 3-base pair deletion
E. Silent mutation
F. Trinucleotide repeat expansion
G. Missense mutation

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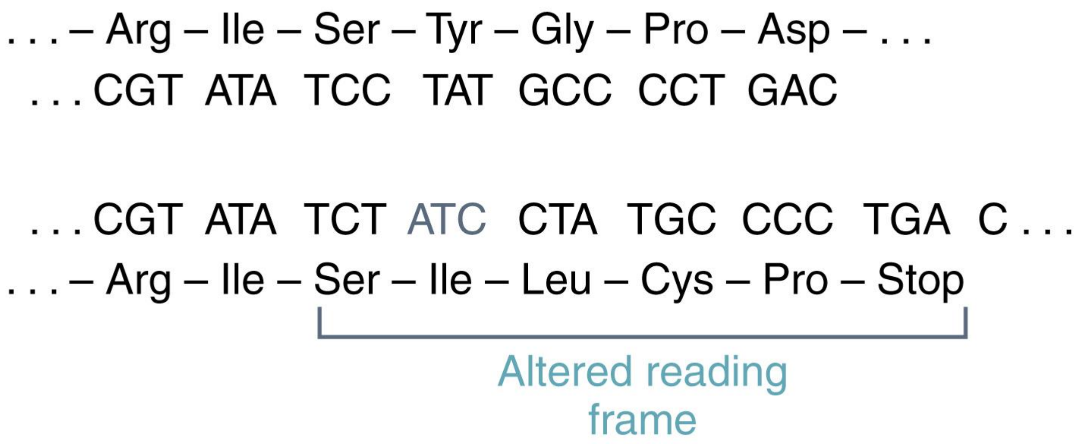

A. Point mutation
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C. Frame-shift mutation
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E. Silent mutation
F. Trinucleotide repeat expansion
G. Missense mutation

SOM.MK.I.BPM1.1.FTM.3.BCHM.0065; Define the terms: reading frame and frame-shift mutation.

A 10-year-old girl is brought to the emergency department by her parents. She has had a 2-day history of progressive weakness with shortness of breath and cough productive of yellow sputum. Her pulse is 115/min, respirations are 26/min, and temperature is 39.7°C (103.4°F). Pulse oximetry on room air shows an oxygen saturation of 92%. Her respirations are shallow and regular. Auscultation shows crackles in the left lower lung lobe. Laboratory studies shows: Hemoglobin: 11 g/dl (12-16), hematocrit: 40% (36-46%), WBC: 16,109/mm^3 (4500-11000), neutrophils 80% (54-62%). She is allergic to penicillin and is prescribed a medication that inhibits bacterial elongation by blocking ribosome translocation. Which of the following medications was most likely prescribed for this patient?

A. Tetracycline
B. Erythromycin
C. Cycloheximide
D. Streptomycin
E. Puromycin
F. Rifampicin
G. Tenofovir
H. Azidothymidine

A 10-year-old girl is brought to the emergency department by her parents. She has had a 2-day history of progressive weakness with shortness of breath and cough productive of yellow sputum. Her pulse is 115/min, respirations are 26/min, and temperature is 39.7°C (103.4°F). Pulse oximetry on room air shows an oxygen saturation of 92%. Her respirations are shallow and regular. Auscultation shows crackles in the left lower lung lobe. Laboratory studies shows: Hemoglobin: 11 g/dl (12-16), hematocrit: 40% (36-46%), WBC: 16,109/mm^3 (4500-11000), neutrophils 80% (54-62%). She is allergic to penicillin and is prescribed a medication that inhibits bacterial elongation by blocking ribosome translocation. Which of the following medications was most likely prescribed for this patient?

A. Tetracycline
✓ B. Erythromycin
C. Cycloheximide
D. Streptomycin
E. Puromycin
F. Rifampicin
G. Tenofovir
H. Azidothymidine

SOM.MK.I.BPM1.1.FTM.3.BCHM.0069; Explain the mode of action of common antibiotics that interfere with translation: Initiation inhibitors (streptomycin), Elongation inhibitors (Tetracycline, chloramphenicol, erythromycin, puromycin, cycloheximide).